

DEM 601 Introduction to Demographic Analysis**Formula sheet**

COVERS

TERM

weeks 1 to 8

Fall 2026

Every equation the course has shown you, in the order it was shown. Each is read out of the deck it appears on, so this sheet cannot fall behind the lectures. The exams are closed book and this is not permitted in them: write what each symbol means beside it, in your own words, because that is the part that is examined.

Week 1 • Introduction to the Course and Basic Demographic Concepts

$$P_{n+k} = P_n + \underbrace{{}_k B_n - {}_k D_n}_{\text{natural increase}} + \underbrace{{}_k I_n - {}_k E_n}_{\text{net migration}}$$

What makes a population change

Week 2 • Demographic Indicators and Person-Years-Lived

$$\text{indicator} = \frac{\text{numerator}}{\text{denominator}} \times k$$

What an indicator is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The mean

$$M = L_0 + \left(\frac{P/2 - \sum f_0}{f_0} \right) i$$

The median of grouped data

$$\text{ratio} = \frac{a}{b} \times k$$

Ratio

$$\text{proportion} = \frac{a}{a+b}$$

Proportion

$$\text{percentage} = \frac{a}{a+b} \times 100$$

Percentage

$$\text{rate} = \frac{e}{PY} \times k$$

A rate needs time

$$CBR = \frac{\text{births in the year}}{\text{mid-year population}} \times 1000$$

Is it a true rate?

$$IMR = \frac{\text{deaths under age 1}}{\text{live births in the year}} \times 1000$$

Is it a true rate?

$$q = \frac{e}{P_0}$$

A rate is not a probability

$$\bar{P} = \frac{P_0 + P_1}{2}$$

The mid-year population

Week 3 • Population Change

$$\Delta P = P_t - P_0$$

Absolute change

$$\% \Delta P = \frac{P_t - P_0}{P_0} \times 100$$

Percentage change

$$P_n = P_0 + (n \times (r \times P_0))$$

Arithmetic growth

$$r = \frac{P_n - P_0}{n} \div P_0 \times 100$$

Arithmetic growth

$$P_n = P_0 \times (1 + r)^n$$

Geometric growth

$$r = \sqrt[n]{\frac{P_n}{P_0}} - 1$$

Geometric growth

$$P_n = P_0 \times e^{r \times n}$$

Exponential growth

$$r = \frac{\ln(P_n/P_0)}{n}$$

Exponential growth

$$r(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta N(t)}{N(t) \Delta t} = \frac{\frac{dN(t)}{dt}}{N(t)} = \frac{d \ln[N(t)]}{dt}$$

Exponential growth

$$P_n = P_0 \times e^{r \times n}$$

Running the formula forwards

$$2P_0 = P_0 \times e^{r \times n}$$

How long until twice as many?

$$\ln(2) = r \times n$$

How long until twice as many?

$$n = \frac{\ln(2)}{r}$$

How long until twice as many?

Week 4 • Age and Sex Composition

$$SR = \frac{P^m}{P^f} \times 100$$

Men per hundred women

$$SR_x = \underbrace{SRB}_{\text{where the cohort began}} \times \underbrace{\frac{S_x^m}{S_x^f}}_{\text{surviving to age } x} \times \underbrace{M_x}_{\text{net migration}}$$

Three things move it

$$f = \frac{\text{total}}{\text{of known age}}$$

Putting back the ages nobody wrote down

$$W = 100 \times \frac{5 (P_{25} + P_{30} + \dots + P_{60})}{P_{23} + P_{24} + \dots + P_{62}}$$

Whipple's index

$$ID = \frac{1}{2} \sum_i |a_i - b_i|$$

The index of dissimilarity

Week 5 • Measures of Mortality

$$CDR = \frac{D}{PY} \times 1000$$

Deaths per thousand people

$${}_n m_x = \frac{{}_n D_x}{{}_n PY_x} \times 1000$$

The rate, one age at a time

$$q_x = \frac{2m_x}{2 + m_x} \quad {}_n q_x = \frac{n \cdot {}_n m_x}{1 + \frac{n}{2} \cdot {}_n m_x}$$

From one to the other, and what it costs

$$\text{share}_c = \frac{D_c}{D} \times 100 \quad \text{rate}_c = \frac{D_c}{PY} \times 100,000$$

A share is not a risk

Week 6 • Standardization of Death Rates & Concepts in Early Life Mortality

$$\text{rate ratio} = \frac{{}_n m_x^A}{{}_n m_x^B} \times 100$$

The comparison that needs no repair

$$CDR = \sum_x {}_n m_x \cdot {}_n c_x$$

A summary rate is rates times weights

$${}_n m_x = \frac{{}_n D_x}{{}_n P_x} \quad {}_n c_x = \frac{{}_n P_x}{P}, \quad \sum_x {}_n c_x = 1$$

A summary rate is rates times weights

$$M = \sum_c M_c \cdot C_c$$

Any summary measure, any composition

$$M_c \text{ is } M \text{ within category } c \text{ alone} \quad C_c = \frac{n_c}{n}, \quad \sum_c C_c = 1$$

Any summary measure, any composition

$$\text{ASDR}^j = \sum_x {}_n m_x^j \cdot {}_n c_x^s \quad {}_n m_x^j \text{ is population } j\text{'s own rate, } {}_n c_x^s \text{ the standard's share}$$

Somebody else's people, your own rates

$$\text{CDR}^A - \text{CDR}^B = \sum_x ({}_n m_x^A - {}_n m_x^B) \bar{c}_x + \sum_x ({}_n c_x^A - {}_n c_x^B) \bar{m}_x$$

Not repairing a difference, splitting it

$$\bar{c}_x = \frac{1}{2} ({}_n c_x^A + {}_n c_x^B) \quad \bar{m}_x = \frac{1}{2} ({}_n m_x^A + {}_n m_x^B)$$

Not repairing a difference, splitting it

$$\text{SMR} = \frac{D^j}{\sum_x {}_n m_x^s \cdot {}_n P_x^j} \times 100 \quad \text{ISDR} = \frac{\text{SMR}}{100} \times \text{CDR}^s$$

Your own people, somebody else's rates

$$\text{IMR} = \frac{\text{deaths under age 1 in year } t}{\text{live births in year } t} \times 1000$$

Not a population, a cohort of births

Week 8 • Life Table I

$${}_n p_x = \frac{\ell_{x+n}}{\ell_x} \quad {}_n q_x = 1 - {}_n p_x = \frac{{}_n d_x}{\ell_x}$$

Who is left at an exact age

$${}_n L_x = n \cdot \ell_{x+n} + {}_n a_x \cdot {}_n d_x$$

The years lived inside one interval

$$T_x = \sum_{a \geq x} {}_nL_a \qquad e_x = \frac{T_x}{\ell_x}$$

The only column anybody ever quotes

$$P_x = B \frac{{}_nL_x}{\ell_0} \qquad P = B e_0 \qquad CDR = \frac{1000}{e_0}$$

The same column, read as a population

$${}_1a_0 = \alpha + \beta m_0$$

Where our own a-nought came from

$$T_x = {}_\infty L_x = \frac{\ell_x}{{}_\infty m_x} \qquad e_x = \frac{1}{{}_\infty m_x}$$

The row with no width

$${}_nq_x = \frac{{}_n \cdot {}_n m_x}{1 + ({}_n - {}_n a_x) \cdot {}_n m_x}$$

Why the rate is not the probability